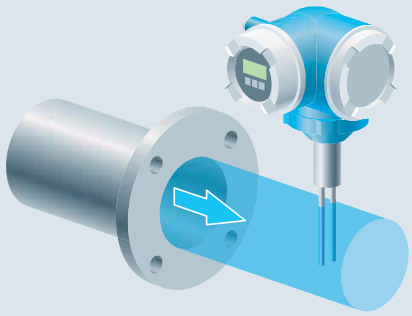
Thermal mass flowmeters

Proline t-mass – For direct mass measurement of industrial gases, compressed air and aqueous fluids even at the lowest flow velocities and pressures



Measuring principle

Many people are made uncomfortably cold by just a small draft. The thermal flow measuring principle is based on the fact that heat is drawn from a heated body when a fluid flows past.
A thermal flowmeter contains two PT100 temperature sensors for this purpose. One sensor measures the current fluid temperature as a reference. The second sensor is heated and has a constant temperature differential relative to the first sensor at "zero flow."
As soon as the fluid begins to flow in the measuring tube, the heated temperature sensor cools off due to the fluid flowing past – the higher the flow velocity, the greater the cooling effect.
The electric current required to maintain the temperature differential is thus a direct measure of mass flow.

Advantages at a glance

- Multivariable – direct measurement and display of mass flow and fluid temperature
- No pressure or temperature compensation required
- High turndown (max. up to 1000:1)
- Excellent low-end sensitivity
- Quick reaction to fluctuations in flow
- Negligible pressure loss
- Maintenance-free, no moving parts

Measuring principle movie:
www.eh.digital/3m2U7TP





The thermal measuring principle is widespread in industry and is being used successfully in many applications with gas flow, for example:

- Compressed air (consumption, distribution)
- Carbon dioxide (beverage production, chilling)
- Argon (steel production)
- Nitrogen and oxygen (production)
- Natural gas (burner, boiler feed control)
- Air and biogas measurement (wastewater plants)

Whenever high turndown or low pressure losses are important in gas metering applications, thermal mass flowmeters offer a real alternative to traditional measuring techniques – whether for process control, consumption and supply monitoring, detecting leaks or monitoring distribution networks. Using insertion versions, it is also possible to measure gas flows in very large pipelines or in rectangular ducts.



t-mass sensors

For basic gas applications

t-mass A
Inline version

- Max. measured error:
  ±3% o.r. (15 to 100% o.f.s.)
  ±0.45% o.f.s. (1 to 15% o.f.s.)
- Process pressure:
  PN 10 to 40 (Class 150 to 300)
- −40 to +100 °C
  (−40 to +212 °F)
- DN 15 to 50 (½ to 2")



t-mass B
Insertion version

- Suitable for large pipelines and rectangular ducts
- Max. measured error:
  ±3% o.r. (15 to 100% o.f.s.)
  ±0.45% o.f.s. (1 to 15% o.f.s.)
- Process pressure: −0.5 to +20 barg
  (7 to 290 psig)
- −40 to +100 °C (−40 to +212 °F)
- DN 80 to 1500 (3 to 60")



For demanding gas applications

t-mass F
Inline version

- Max. measured error:
  ±1% o.r. (10 to 100% o.f.s.)
  ±0.1% o.f.s. (1 to 10% o.f.s.)
- Process pressure: Up to PN 40
  (up to Class 300)
- −40 to +100 °C
  (−40 to +212 °F)
- DN 15 to 100 (½ to 4")



F L **E** X

t-mass I
Insertion version

- Suitable for large pipelines and rectangular ducts
- Max. measured error:
  ±1% o.r. (10 to 100% o.f.s.)
  ±0.1% o.f.s. (1 to 10% o.f.s.)
- Process pressure: up to 40 barg
  (580 psig)
- −40 to +130 °C (−40 to +266 °F)
- DN 80 to 1500 (3 to 60")



F L **E** X

Flexible installation

Whether in rectangular ventilation ducts or in pipes – t-mass sensors are always the perfect fit. For installation, both inline and insertion versions in various nominal diameters are available:

Inline version

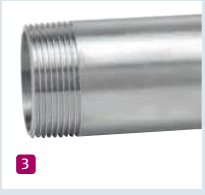
1. With flange (t-mass A and F)
2. With lap-joint flange (t-mass A)
3. With external thread (t-mass A and F)

Insertion version

4. Suitable for pipelines or rectangular ventilation ducts up to DN 1500 (60")
5. Optionally with "hot tap" mounting tool for inserting or removing the sensor under operating conditions:
   – For recalibration
   – For certification
   – For service purposes
   – For mobile use











Proline transmitters

t-mass sensors	Proline transmitters		
	150	300	500
			
t-mass A	✔	–	–
t-mass B	✔	–	–
t-mass F	–	✔	✔
t-mass I	–	✔	✔
t-mass T	✔	–	–

Detailed description of the Proline 300/500 transmitters ▸ page 9 to 11

Proline 150

For basic applications (cost-effective measurement)

- For t-mass A, B and T sensors
- Easy-to-understand, four-line display with three push buttons
- Display/outputs for flow and temperature
- Free selection of up to 4 fluids
- Switch and/or relay output for alarm messages
- Totalizer functions



Gas Engine

Gas programming made easy

Depending on pressure and temperature, gases change their volume and their specific properties such as operating density, heat capacity or viscosity. Converting operating volume to standard volume is highly labor-intensive and inconvenient.
With the t-mass transmitter's integrated "Gas Engine" function, and an automatic temperature and pressure compensation, gases and gas mixtures can now be measured highly reliably:

- 22 freely selectable gases (4 gases for Proline 150)
- 2 gas mixtures with up to 8 user-definable components (Proline 300/500)
- Toggling between 2 gas mixtures (Proline 300/500)
- Programmed gases can be changed at any time (without recalibration)

t-mass T 150

For simple liquid flow monitoring

Many industrial processes and utilities require reliable measurement, monitoring and control of liquid flows. Proline t-mass T 150 works according to the thermal measuring principle and measures all aqueous fluids, for example cooling water, heating water, demineralized water (condensate), industrial water, WFI water, irrigation water or fluids in the wastewater area:

- Independent of the electrical conductivity
- Fast commissioning via the local display
- Compact insertion version (DN 40 to 1000/1½ to 40")
- Standard or hygienic version (3-A, EHEDG)
- Maintenance-free, no moving parts
- Wide variety of applications: monitoring, flow measurement, switch function, etc.



